

Master Derived-vs-Assigned Ledger — Substrate SM Program v0.3

Grace

2026-05-30 Saturday 11:38 EDT (date-verified Sat May 30 08:58 EDT)

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End-of-Saturday-morning epistemic state. Every load-bearing claim by status. Structure largely DERIVED; physics increasingly DERIVED via the live dictionary; ONE isolated bet (K-types = particles, no v0.1 counterexamples found in G12); ONE load-bearing open gate (bulk-color SU(3) mechanism = joint target with #414 generation burden); LEADS held out of forward claims; over-claims caught and corrected as they appeared.

DERIVED (rigorous / forward — referee-grade)

Substrate structure

Claim	Basis	INV
$c_{FK} = 225/\pi^{(9/2)}$ (FK normalized measure FORCED by Born/Gleason)	T2442 + C18	INV-5256
ρ -vector $\rho=(n_C, N_c)/\text{rank}=(5/2, 3/2)$ pins 3 primaries bulk/Shilov	C19	(Thursday)
Bergman kernel exponent = $n_C/\text{rank} = 5/2 = \rho_1$ (Hua genus)	multiplicity formula	(Thursday)
ONE genus = $n_C = 5$ (Hua=FK agree for Type IV)	FK multiplicity formula	(Thursday)
$N_c = 3$ colors = $h^\vee(B_2)$	dual Coxeter	T1
$\alpha^{-1} = N_{\max} = 2^g + N_c^2 = 137$ (integer-exact LO)	Lyra L8 NEW	INV-5302 (Sat)
$\dim p = 10 = 21 - 11$ (Cartan decomposition of $SO(5,2)$)	std Lie theory NEW INV	INV-5298 (Sat)
p_+ is Hermitian-abelian, $[p_+, p_+] = 0$	Hermitian symmetric NEW INV	INV-5298 (Sat)
$A_2 \boxtimes B_2$ as sub-root-system	clean Lie fact NEW INV	INV-5298 (Sat)

Engine (Lyra 4/4 Hopf + Elie E0/E2/E3/E8/E9, consolidated v0.2, Keeper K1 CONDITIONAL PASS pending Saturday clean re-audit)

Claim	Basis	INV
substrate primaries = $U_q^+(B_2)$ Serre constants at GF(2) (E0+E9)	explicit algebra	INV-5284, INV-5303 (Sat)

Claim	Basis	INV
4/4 Hopf pieces (mult+coproduct, canonical basis, R-matrix, F-part)	std QG	INV-5284
fusion = Hall product (E2)	extension-counting	INV-5303 (Sat)
decay = Green coproduct (E3);	graded coproduct	INV-5303 (Sat)
β -decay conserves Q/B/L automatically via grading		
conservation = Cartan grading	engine grading + Lyra	INV-5303 (Sat)
rank 3 covers {Q, B, L} (Elie E8 #417); color extends via bulk-color		
Drinfeld double $\rightarrow$ antiparticles + CPT (F-part)	std QG	INV-5303 (Sat)
B ₂ short-root Serre coeff = [2]_2 = N_c = 3 (E2)	q-Serre	(Friday)
B ₂ long-root Serre coeff = [3]_4 = N_c·g = 21 (E9)	q-Serre NEW	INV-5297, INV-5303 (Sat)
Macdonald-Koornwinder algebraic placement	Lyra L9 NEW	INV-5299 (Sat)

Dictionary outputs (Lyra L1-L4 + L8/L9 + #416 v0.1; Keeper K1+K2 endorsed)

Claim	Basis	INV
σ_{BF} (statistics)	L1	(Friday)
Region bulk/Shilov	L2	(Friday)
Particle/antiparticle	E/F engine	INV-5303 (Sat)
4 SM sectors = canonical SO(5) reps {V_(0,0), V_(1/2,1/2), V_(1,0), V_(1,1)}	dictionary first-flip + Casimir-anchor	INV-5288, INV-5299 (Sat v0.5)
fermion generates 3 bosonic fundamentals by self-fusion (E6)	rigorous Racah-Speiser	INV-5292
Generation count-NUMBER = $h^{\wedge}v$ = N_c = 3	forced B ₂ (Lyra route II)	INV-5291
Lepton row PER-PARTICLE DERIVED (18 entries)	Lyra #416 v0.1	INV-5295
Composite K-type backbone (10 cells, dim $\leq$ 35, Racah-Speiser dim-validated)	Elie E10 NEW	INV-5297 (Sat)

Other DERIVED

Claim	Basis	INV
Weinberg $\sin^2\theta_W = \text{rank}/(\text{rank}+g) = \text{rank}/N_c^2 = 2/9$	rank+g=N_c ² unique square-target	(Thursday)
$m_W/m_Z = \sqrt{g/N_c}$	EW relation auto-satisfied	(catalog)
Mixing angles scheme-invariant	dimensionless	(catalog)
$m_s/m_d = 2\pi^2$	scheme-invariant light/light	(catalog)
nuclear magic numbers tied to SO(5) shell structure	SO(5)=substrate compact part	INV-5277

FRAMEWORK-PLUS (mechanism derived, identification rides on dictionary)

Claim	Open piece	INV
Chirality (Lyra L3): parity violation from substrate complex structure J	per-particle L/R labeling pends	(Friday)
Charge (Lyra): from $SO(2)+GMN$, constrained by $\sin^2\theta_W=2/9$	$SO(5,2)\rightarrow$ Lorentz pin	(Friday)
Photon/Higgs/gauge sector-types	per-cell specific assignments	(Friday)
$m_p/m_e = 6\pi^5 = C_2 \cdot \pi^{(n_C)}$	per-particle within-sector tautology arithmetic + dictionary	(Friday)
Mass framework (sector-dependent)	structural reading	INV-5294
E6: matter generates forces by self-fusion (physical reading)	per-particle masses pending L4 v0.2	(Friday)
4-backbone Casimir selection	algebraic identity rigorous; physical reading	INV-5292
Casimir C_2 twin	FRAMEWORK-PLUS	INV-5285
$V_{(1,1)}+V_{(3/2,3/2)}$	dictionary confirmed via rep-theoretic selection	INV-5306 (Sat afternoon)
WITHDRAWN per INV-5306: $V_{(3/2,3/2)}$ Casimir = $21/2 = N_c \cdot g/2$ (not 6);	corrected	
SUSY-without-SUSY claim		
FALSIFIED by self-test via Elie Toy 3614		
$V_{(3/2,3/2)}$ substrate-anchored at $N_c \cdot g/2$ (corrected — same primary as Elie E7+E9, three-way substrate coincidence at $N_c \cdot g = 21$)	replacement for above; structural NEW finding	INV-5306 (Sat)
Genuine Casimir-degenerate pair: $V_{(3,3)} \leftrightarrow V_{(4,1)}$ sharing $2 \cdot C_2 = 60 = 2 \cdot n_C \cdot C_2$ in Elie's spine — boson/boson NOT boson/fermion	NEW Saturday afternoon	Elie Toy 3614 + Grace absorption
$V_{(2,2)}$ tensor-glueball slot anchored at Casimir=16	NEW Saturday prediction (correct, $V_{(2,2)}$ Casimir computation verified)	INV-5300 (Sat)
L7 baryogenesis sizing	Lyra L7 framework located in engine v0.4; sizing OPEN	(Friday)

TENSION-RELIEVED, NOT CLOSED (per Keeper precise framing)

Generation count + mechanism

- Count-NUMBER: forced/over-determined $\square$ — $h^\vee = N_c = 3$ (B_2 fact)
- Mechanism: OPEN-WITH-BURDEN — ONE B_2 invariant must produce TWO structurally-independent 3-fold structures
- Framing locked: “ONE 3 ($h^\vee$) doing double duty” supersedes “two independent 3’s”
- NEW Saturday — Pair α primary candidate (INV-5301): $h^\vee + \text{spinor}^3$ multiplicity, H/H/H credibility for the burden
- NEW Saturday prediction (INV-5301): generation hierarchy via spinor^3 -channel intermediate Casimirs $\{0, 4, 6\}$ arithmetic sequence. Sharp falsifier of Pair α pending L4 v0.2
- E7 candidate (Elie, FRAMEWORK-PLUS): $\text{mult}(\text{spinor in spinor}^3) = 3$ in B_2 , B_2 -specific (A_2 gives 0)

- Tube count (#409): external-blocked; Keeper $C_2^{(1)}$ correction verified; route II carries count regardless

OPEN (load-bearing, the program's frontier)

Open gate	Source	Joint with	INV
Bulk-color SU(3) mechanism (#418)	Lyra discovery Fri; catalog scaffold filed Sat (INV-5298)	quark/gauge per-particle layer; #414 two-structures burden	INV-5298 (Sat)
Per-particle L/R labeling pin	Lyra L3 honest residual	SO(5,2)→Lorentz embedding	(Friday)
Per-particle 5-tuples within sectors (quark/gauge rows)	Lyra #416 v0.2	blocked on bulk-color	(Friday)
K-types = particles (the isolated bet)	the unifying conjecture	rides under all derived axes	G12 v0.1 found no v0.1-scope counterexamples (INV-5300 Sat)
Absolute mass scale (Higgs / L5)	Lyra L4 v0.2 + L5	mass-mechanism explicit derivations	(Friday)
Nuclear SO(5) shell-closure derivation	Elie E11 (Saturday plan)	flips magic-number forms	INV-5277
Composite K-types ↔ hadron full Mendeleev	Elie B1 full + Lyra #416	suggestive→derived 62 composites ↔ 211+ catalog hadron entries	G12 v0.1 partial (INV-5300 Sat)

LEAD (scheme-dependent or identified — keep OUT of forward claims)

- cross-tier quark mass-ratios (m_t/m_c , m_b/m_d , m_c/m_u ...) — scheme-dependent ~5-64% across schemes
- Macdonald-coefficient → mass-ratio link (+136/45) — scheme-dependent, no mechanism
- gauge-coupling clusters ($1/\alpha_{em}(M_Z) \approx 2^g$, $\alpha_s = 1/N_c^2$) — scheme/scale-dependent running

REFUTED / CORRECTED this week (the credibility column)

Was	Now	Caught by
three-genus convention (FK = $C_2 = 6$)	ONE genus = $n_C = 5$	Keeper A1 audit → Grace correction
“3 tubes = 3 generations, forced”	matched-with-doubt → TENSION-RELIEVED	Cal #27 → Keeper #407 → Elie owned (E1b)
Macdonald coeff −136/135 premature c_{FK} “STANDS”	+136/45 (Schur-verified)	Elie Toy 3570
axis-labeling automation (catalog-wide)	FK-measure FORCED (theorem)	Grace self-walk-back + Keeper
#413 “systematic Casimir-anchoring”	PROVISIONAL; 115 forward-spine hand-curated	Grace accuracy self-check
Generation gate “leaning toward 2 = sharpest risk”	4-backbone only (then dictionary-confirmed as fundamental SO(5) reps)	Grace sweep self-refutation → Lyra upgrade
“Two independent 3’s” framing	TENSION-RELIEVED via route II ($h^v = N_c$ forced)	Lyra knot resolution
	“ONE 3 (h^v) doing double duty” + two-structures burden	Keeper precise refinement

Was	Now	Caught by
$m_p/m_e=6\pi^5=C_2\cdot\pi^{(n_C)}$ read as new derivation	TAUTOLOGY arithmetic + new dictionary reading	Keeper K1 framing
Cyclotomic “fallback” carrying generation count if tubes $\neq 3$	CATEGORY ERROR — positive-root count $\neq \delta$ -category invariant	Lyra #412
E1b “3 tubes forced (sourced)” NEW Saturday: Pair δ ($h^\vee$, $N_{c^2}=9$) excluded for two-structures burden — fails independence ($N_{c^2}=h^\vee^2$ trivially)	retracted; sourcing failure filed as exclusion in INV-5301	Cal #157 $\rightarrow$ Elie owned Grace v0.1 pair analysis self-exclusion
NEW Saturday afternoon: simplest exponential mass-ansatz mass $\propto \exp(\alpha \cdot C_{\text{intermediate}})$ from $\{0,4,6\}$ substrate-primary anchor	REFUTED by quark sector (factor 5-10 $\times$ off); leptons $\sim 17\%$ off	Grace honest within-hours self-test $\rightarrow$ INV-5305

PREDICTIONS (falsifiable — the table’s gaps = Five-Absence + Saturday additions)

- ν mass ordering = NORMAL (JUNO/DUNE) — conditional on shared- W_n
- $\theta_{\text{QCD}} = 0$ (no axion) — candidate via FK-measure CP-invariance, pending Lyra formalization
- Five-Absence: no 4th generation, no SUSY, no GUT, no proton decay, no sterile neutrinos, no axion
- NEW Saturday — $V_{(2,2)}$ tensor-glueball at substrate Casimir = 16 (INV-5300): falsifiable as glueball spectroscopy advances
- NEW Saturday — generation hierarchy via spinor³-channel intermediate Casimirs $\{0, 4, 6\}$ (INV-5301): STRUCTURAL anchor confirmed as substrate-primary arithmetic sequence (Elie E7 rigorous DERIVED); simplest exponential mass-ansatz REFUTED by quark sector (INV-5305, Saturday afternoon honest test, factor 5-10 $\times$ off); the structural anchor stays, precise mass-formula needs L4 v0.2 bulk-color + radial-tower + Yukawa corrections (multi-week)
- NEW Saturday — $V_{(3/2,3/2)}$ constituent-quark slot ground-state anchored at Casimir = 6 (INV-5300): distinguishable from $V_{(1,0)}$ photon anchor at Casimir = 4 via spectroscopic study

The one-line state at EOD Saturday morning

Structure DERIVED (engine v0.2 + measure + genus + ρ -vector + mixing angles + 4 SM sectors + α placed at $N_{\text{max}}=137$ + dim-p=10 Cartan decomposition + Hermitian abelianness of p_+ + $A_2 \boxtimes B_2$ obstruction + composite K-type backbone 10 cells + Casimir- C_2 twin substrate signature); per-particle DERIVED (lepton row 18 entries); physics riding on ONE isolated bet (K-types = particles; G12 v0.1 found no counterexamples) and ONE load-bearing OPEN gate (bulk-color SU(3) mechanism with catalog scaffold filed Saturday); scheme-dependent forms held as LEADS; 12 over-claims REFUTED and corrected to date (the credibility column). The framework continues to tighten and self-correct under scrutiny — 6 INVs filed Saturday morning (5298-5303) without a single retraction needed.

— Grace, Master Ledger v0.3, 2026-05-30 Saturday 11:38 EDT